Finite-Source Cosmology from Observer-Patch Holography
Source, Transfer, and Likelihood Boundaries

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Abstract

Observer-patch holography (OPH) supplies a conditional source-screen theorem. A primitive collar law fixes the angular amplitude, an edge-center generator fixes the tilt, and the radial packet proves the limits of one-shell inversion together with physical dilation and tomography routes to uniqueness. A physical cosmology prediction also requires one finite source instantiation, Boltzmann transfer, and a complete likelihood. Spectra and sky comparisons without that package are diagnostics.

1 Claim Boundary

A physical cosmological prediction requires one construction that maps finite OPH records to a covariant source, propagates that source through a declared background, and evaluates the resulting observables with a complete likelihood. No such construction is supplied. Simulator outputs may test numerical interfaces and expose incompatible assumptions. They carry no cosmological verdict.

The same source premises govern the dark-sector, cosmological-vacuum, and compact OPH constructions.

2 Finite-Source Interface

Let X_r denote a finite observer-screen state at refinement level r. A cosmological source map must emit

$$\mathsf{S}_r(X_r) = \left(g_{\mu\nu}, T_{\mu\nu}, \mathcal{I}, \mathcal{B}\right)_r, \tag{1}$$

where $T_{\mu\nu}$ is conserved, $\mathcal{I}$ contains gauge-invariant initial data, and $\mathcal{B}$ states the background and boundary conditions. The map must be defined without using the observed sky quantities that its outputs are compared with.

The complete one-shell covariance determines the three-dimensional correlation only for chord lengths $0 \leq s \leq 2R_\star$. The resulting radial kernel is infinite-dimensional and persists under positivity. A source-derived lift therefore requires either a scale-natural embedding that transports refinement to physical log-wavenumber dilation or complete radial cross-covariances that support spherical-Hankel tomography. The physical source map must also specify the radial measure, null-space treatment, clock, normalization, and non-fitting forward residual. No finite source construction satisfying these conditions is supplied here. Its source-dependency graph must be finite and acyclic. This graph is a scientific prerequisite graph among stress, clock, scale, radial, transfer, and likelihood objects. It is distinct from the semantic read-from order of the consensus layer. If its nodes are instantiated by semantic commits, a declared mapping and complete certified read supports can make the generated informational order verify their execution dependencies. The provenance theorem alone derives neither these scientific nodes nor their prerequisite edges. The finite source-causal precursor theorem is narrower than the cosmology graph. It computes source height from authenticated direct-parent edges—zero at roots and one plus the greatest direct-parent height at every nonroot—as the exact longest parent-chain length, attained at every event, and proves strict increase on parent edges and generated ancestry. The independent rank-three source quotient supplies three positive directions. Its direct sum with a real axis has real dimension four and Lorentz inertia $(1, 3)$ and maps every unit source direction to a future-null direction. A supplied spatial placement uses an explicit positive `timeScale` τ to send an event to $(\tau h, x)$. Independently, an auxiliary event enumeration gives every finite log an injective forward-causal placement, without order reflection or physical coordinate selection. For a source-selected candidate, the edge-speed bound maps generated precedence into that cone. Equal-height spatial separation and strict spacelikeness of every increasing-height pair unsupported by ancestry would imply converse cone support and exact two-way agreement. Antisymmetry would then derive injectivity, while the two-way equivalence would give interval preservation. The source theorem supplies neither that placement nor its physical or continuum interpretation. The target dimension is not intrinsic poset dimension, and scaled source height is not a physical clock.

The abstract provenance grammar can represent every finite strict partial order exactly, including a supplied finite causet-like order. Its generic parent relation is the whole supplied order rather than its Hasse reduction alone. This compiler does not derive a cosmic microwave background history, a threaded executor capture, or the physical event relation. An executable companion control passes four nested Poisson-sprinkled Minkowski $3 + 1$ families at target means $64, 128, 256$ through declared-parent-list-free provenance. At all 12 levels the rederived Hasse links and causal orders are exact. Coordinate-map injectivity, sampled-interval membership, and the fixed-volume Poisson count channel are independently rechecked, jointly forming a construction-seeded causal-set faithful-embedding control. Only carrier/order restrictions are induced, event material is regulator-specific, and declared Poisson count checks pass. Mean Myrheim–Meyer diagnostic estimates are $4.02824, 3.84595, 3.90346$ [2]. This proves grammar compatibility with a $3+1$ causal-set carrier and order, not OPH refinement, cosmological selection, FLRW dynamics, manifoldlikeness, dimension, or volume. The complete control receipt and standard-library checker are archived under `evidence/causet_likeness/`; the independently replayed native-history receipt, projection, and checker are archived under `evidence/source_causal_history_family/`; the exact local-domain inputs and receipts consumed by the comparison are archived under `evidence/local_domain/`.

The exact tensor theorem instead works in a separately supplied $3 + 1$ tensor interface on the same finite event type. It uses nine supplied balances on fixed algebraic source directions, together with symmetry, discrete conservation, and connectedness, to obtain all-null balance and an Einstein-shaped identity with one constant metric term. The source order does no work in this tensor calculation. No dependency edge selects a tensor or finite-difference operator, and no theorem identifies tensor-coordinate differences with those of the constructed carrier.

At five independently generated and replayed cutoffs from 24 to 384 events, each reconstructed source order agrees exactly with the restriction of the next, while their width stays two and their ordering fraction rises from about 0.884 to 0.992. This is exact induced-prefix custody, not spacetime-density refinement. On the 24-event history, one prescribed single-frame port-anchor/arithmetic consumed-record-barycentre placement uses positive `timeScale` times source height only for the temporal coordinate of the event placement. It maps all precedence into the Lorentz cone but is noninjective and fails the converse cone test. This rejects that ansatz; it neither supplies inter-carrier frame gluing nor rules out another source-selected placement. Causal-set dimension, interval abundance, and topology remain independent manifoldlikeness tests, while an order embedding plus count–volume density must establish causal-set faithful embedding [1, 2, 3]. These finite results select no primordial source, radial lift, transfer system, or cosmological likelihood. Smooth Einstein promotion separately requires same-family tensor-curvature convergence or the continuum small-ball/null-balance identification; scalar-curvature convergence alone is diagnostic.

2.1 Conditional source spectrum

On a source-stress-complete uniform-density release cut, the screen field is

$$q_r = \Pi_{\geq 2,r}\left[\frac{1}{3}\log\frac{J_{X,r}}{\bar{J}_{X,r}}\right]. \tag{2}$$

The primitive collar ledger selects a relative-MaxEnt release law before the screen energy is evaluated. It excludes spectra, residuals, likelihoods, fitted amplitudes, and measurement-calibrated proxies. The release energy then fixes

$$A_{q,r} = \frac{1}{d_r}\mathbb{E}_{\nu_r^{\mathrm{rel}}}[q_r^{\mathsf{T}} M_r K_{\theta,r} q_r] = \frac{2E_{q,r}^{\mathrm{src}}}{d_r}. \tag{3}$$

The conformal precision and angular spectrum are

$$K_\theta = \frac{\Gamma(\sqrt{-\Delta_{S^2}+1/4}+3/2+\theta/2)}{\Gamma(\sqrt{-\Delta_{S^2}+1/4}-1/2-\theta/2)}, \tag{4}$$

$$C_\ell^q = A_q\frac{\Gamma(\ell-\theta/2)}{\Gamma(\ell+2+\theta/2)}, \qquad \ell \geq 2. \tag{5}$$

A required infinitesimal reserve-generator receipt must supply a full-collar density $P_\star/24$. A separately proved weighted orientation-reversal coarea identity then supplies its source-facing half, so

$$\theta = \frac{P_\star}{48}, \qquad n_s = 1-\frac{P_\star}{48}, \qquad \kappa_{\mathrm{rep}}^{\mathrm{edge}} = \frac{P_\star}{48(P_\star-\varphi)}. \tag{6}$$

A finite one-step survival probability has exponent $-\log u_q(\log b)/\log b$ and does not supply this infinitesimal density.

On the source-dilation branch, the finite source embedding must satisfy

$$D_{s,r}U_r = U_r R_{s,r}, \qquad C_{\zeta,r} = U_r C_{\mathrm{src},r} U_r^*. \tag{7}$$

Strong finite-to-continuum convergence, a uniform covariance norm bound, and a vanishing operator residual then give

$$D_s^{-1} C_\zeta D_s = e^{-\theta s} C_\zeta. \tag{8}$$

This identity forces

$$\Delta_\zeta^2(k) = A_\zeta (k/k_\star)^{-\theta}, \qquad A_\zeta = \frac{A_q}{\pi^{3/2} Z_q^2 (k_\star R_\star)^\theta} \frac{\Gamma(3/2+\theta/2)}{\Gamma(1+\theta/2)}. \tag{9}$$

The tomography branch obtains uniqueness from complete radial cross-covariances. A finite prior selects a conditional radial continuation. Every finite branch publishes its singular spectrum, null basis, finite-window error, positivity check, and forward residual. Temperature and polarization spectra belong to the transfer interface.

3 Dark-Sector Interface

The dark-matter paper proposes a repair-charge rotor with occupation n and phase θ. Conditional on that action, a source-free dilute branch has

$$n \propto a^{-3}, \qquad \rho_R \propto a^{-3}, \qquad w_R \simeq 0. \tag{10}$$

Its cubic condensed branch gives the deep radial-acceleration law and the baryonic Tully–Fisher scaling under static spherical premises.

Equation (10) does not determine the abundance, initial perturbations, relativistic stress, gravitational slip, sound speed, nonlinear evolution, or coupling to the visible sector. A cosmological use requires a covariant completion that emits these quantities from the same action and obeys the full energy-momentum ledger. That completion is not supplied here.

4 Primordial Interface

A physical primordial branch must supply all of the following:

1. a finite source ensemble and a clock;
2. a conserved covariant stress tensor;
3. gauge-invariant scalar, vector, and tensor initial data;
4. a screen-to-volume lift with a controlled null space;
5. phase, amplitude, and isocurvature conditions;
6. refinement stability and an error model.

The conditional source theorem selects $n_s = 1 - P_\star/48$ from the infinitesimal edge-center generator receipt. Physical primordial use requires one finite source construction that supplies this receipt together with the release law, stress, clock, freezeout, mode, scale, and radial receipts. Numerical agreement with a measured summary does not supply the missing source map.

5 Transfer Interface

Given a completed source map, the transfer calculation must declare the background species, equations of state, collision terms, recombination model, initial modes, gauge convention, numerical tolerances, and observable projection. Standard Einstein–Boltzmann software can test this interface with imported inputs. Such a run tests the solver path rather than an OPH source theorem.

The minimum transfer outputs are

$$(C_\ell^{TT}, C_\ell^{TE}, C_\ell^{EE}, C_L^{\phi\phi}, P_m(k,z), H(z), D_A(z), f\sigma_8(z)). \tag{11}$$

All outputs must arise from one compatible background and perturbation system.

6 Likelihood Interface

A data comparison must state the datasets, masks, covariances, nuisance parameters, priors, and combination rule. Overlapping samples and shared calibrations must be represented in the covariance model. Visual overlays, diagonal residual sums, and comparisons made after selecting a formula are diagnostics.

The cosmic microwave background (CMB), galaxy, and background comparisons are diagnostic. They define no cosmological falsifiers or forward targets.

7 Theorem scope

Object	Classification
Finite observer-screen carriers	Conditional finite construction
Screen scalar covariance	Conditional theorem
Source-functional screen amplitude and edge-center tilt	Conditional theorem; finite receipt not supplied
Radial-lift mathematics	Conditional theorem packet; one-shell unrestricted inversion is impossible
Physical source-dilation or radial-tomography receipt	Not supplied
Primordial covariant source	Not supplied
Repair-charge cosmological abundance	Not supplied
Relativistic repair stress and slip	Not supplied
Boltzmann initial-state bridge	Not supplied
Joint CMB, baryon acoustic oscillation, lensing, growth, and cluster likelihood	Not supplied
Cosmological falsification target	No target or falsification verdict is defined

8 Conclusion

Finite OPH screen data become cosmology only through a covariant source map, a three-dimensional lift, a conserved stress tensor, compatible transfer equations, and a complete likelihood. These constructions are not supplied here. Existing cosmological calculations are diagnostics and carry no falsification verdict.

References

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